

CS-540 Data Literacy & Visualizations

7-1 Submit Activity: Data Exploration and Visualization for Insights

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Transit Ridership Data Analysis Report

The goal of this analysis was to gather relevant ridership data, clean and prepare the dataset for analysis, and generate visualizations using Python to better understand trends and disparities across the selected areas. Three visualizations were created to explore the data from different perspectives: a line chart showing ridership trends over time, a bar chart comparing total ridership across neighborhoods, and a scatterplot examining the relationship between population size and ridership. These visualizations were used to identify patterns in transit usage that could support strategic decision making for organizations seeking to improve public transportation services.

Trends, Differences, and Outliers Observed in the Data

The analysis revealed several trends and differences in transit usage across the three neighborhoods. The line chart showing ridership trends over time illustrated how ridership levels changed across different years for the three agencies. One of the most noticeable trends was that King County consistently showed higher ridership numbers compared to the other two agencies. This is likely due to the larger population and more extensive transit infrastructure present in that region. In contrast, the City of Longview showed significantly lower ridership numbers throughout the dataset. This difference may be explained by the smaller population size and potentially fewer transit routes or services available in that area. Spokane Transit Authority

showed moderate ridership levels, generally falling between King County and Longview in terms of total passenger trips.

Another pattern observed in the data is the relationship between population and ridership levels. The scatterplot comparing service area population to ridership indicates that areas with larger populations tend to have higher transit ridership. This pattern suggests that population density plays an important role in transit demand. However, the relationship is not perfectly linear, indicating that other factors such as service availability, infrastructure, and commuting patterns may also influence ridership levels. No extreme outliers were observed in the dataset, but there were clear differences between the three agencies in terms of scale. King County's ridership numbers were significantly higher than the others, which reflects the differences in population and urbanization among the regions analyzed.

Disparities and Patterns Relevant to the Nonprofit's Goals

Several disparities and patterns identified in the data may be important for a nonprofit organization focused on improving public transportation. The most obvious disparity is the large difference in ridership between King County and the other two agencies. This difference suggests that some areas have significantly more transit usage and potentially better-developed transit networks. Lower ridership levels in areas such as the City of Longview may indicate either lower demand or limited transit accessibility. If the lower ridership is due to insufficient services or limited route coverage, this could highlight an opportunity for targeted improvements. Nonprofits advocating for better transportation systems could use this information to support initiatives that expand transit access in underserved areas. Another

important pattern is the positive relationship between population size and ridership levels. Areas with higher populations naturally tend to generate greater demand for public transportation. However, the scatterplot also suggests that population alone does not fully explain ridership levels. This means that investments in infrastructure, service frequency, and public awareness campaigns may also significantly influence transit usage. Understanding these disparities can help nonprofit organizations identify communities that may benefit most from improvements in transportation access and services.

Summary of What the Data Suggests About Transit Usage Across Neighborhoods

Overall, the data suggest that transit usage varies significantly across the three neighborhoods analyzed. King County shows the highest level of transit ridership, which is consistent with its larger population and more urbanized environment. Spokane Transit Authority demonstrates moderate levels of ridership, while the City of Longview exhibits comparatively lower ridership levels. The visualizations suggest that population size is an important factor influencing ridership levels, but it is not the only determinant. Differences in infrastructure, route coverage, service frequency, and commuting patterns may also play important roles in shaping transit usage across different communities. The findings indicate that transit systems serving larger and more densely populated areas tend to experience higher ridership levels. However, the data also highlights potential opportunities for improving transit accessibility and encouraging greater ridership in smaller or less densely populated regions.

How the Visualizations Support the Conclusions

Each of the three visualizations created in the analysis provided a different perspective on the dataset and helped reveal key insights. The line chart of ridership trends over time allowed for a comparison of how transit usage changed across the three agencies over multiple years. This visualization clearly showed that King County consistently maintained higher ridership levels compared to the other agencies, while Longview remained significantly lower. This trend supports the conclusion that larger urban regions tend to have greater reliance on public transportation. The bar chart comparing total ridership across neighborhoods provided a direct visual comparison of the total ridership numbers between the three agencies. This chart made it easy to identify which neighborhood had the highest and lowest overall ridership levels. The clear differences shown in the bar chart reinforced the observed disparities between the agencies. The scatterplot of ridership versus population helped explore the relationship between service area population and transit usage. The visualization demonstrated a general trend where larger populations correspond to higher ridership levels. This pattern supports the conclusion that population density is a key factor influencing public transportation demand. Together, these visualizations provided multiple ways to interpret the data and strengthened the overall analysis by highlighting patterns that might not be immediately visible in raw numerical tables.

How the Findings Could Inform Nonprofit Decision Making

The insights gained from this analysis could help nonprofit organizations make more informed decisions about where to focus their efforts to improve public transportation systems. For example, the analysis suggests that areas with lower ridership may benefit from targeted

investments in service improvements, infrastructure development, or outreach programs that encourage greater use of transit services. Nonprofits could use this information to advocate for increased funding or policy changes that support improved transit access in communities with limited transportation options. Identifying areas where ridership is lower relative to population size may also help organizations prioritize communities that may be underserved by existing transit networks. Additionally, understanding ridership trends over time can help organizations evaluate whether transportation initiatives or policy changes have had a positive impact on transit usage. This type of analysis can support evidence-based decision-making and strengthen advocacy efforts aimed at improving public transportation.

Importance of Data-Driven Insights in Addressing Real-World Challenges

Data-driven insights are essential for addressing complex real-world challenges such as improving public transportation systems. By analyzing data and identifying patterns, organizations can make more informed decisions that are grounded in evidence rather than assumptions. In the context of public transportation, data analysis can reveal where transit services are most heavily used, where improvements may be needed, and how different factors influence ridership levels. These insights can guide strategic planning, resource allocation, and policy development. For nonprofit organizations, using data to support advocacy efforts can also increase credibility and strengthen proposals for funding or policy changes. Data-driven approaches enable organizations to clearly demonstrate the need for improvements and to measure the effectiveness of initiatives over time. Ultimately, the ability to collect, analyze, and interpret data is a powerful tool for organizations seeking to create meaningful improvements in

public transportation systems and to ensure that communities have reliable and accessible mobility options.